Suryaa Senthilkumar Shanthi

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Summary

Robotics Systems Engineer with hands-on experience building autonomous navigation pipelines, multi-sensor fusion systems, and real-time embedded control on edge compute platforms (Jetson Nano, Raspberry Pi). Proven track record deploying object detection, PID-based motor control, and A* path planning in dynamic environments. Background spans RTL/FPGA design, embedded firmware, and systems-level hardware-software integration.

Education

Georgia Institute of Technology **August 2022 – May 2024**
M.S., Electrical and Computer Engineering (GPA: 3.67/4) *Atlanta, GA*
Adv Computer Architecture, Embedded Systems Design, Adv Programming Techniques, Adv VLSI Systems, Hardware Machine Learning

SRM Institute of Science and Technology **June 2018 – May 2022**
B.Tech., Electronics and Communication Engineering (GPA: 9.24/10) *Chennai, India*
Digital Logic Design, Digital Logic Verification, Linear Integrated Circuits, Verilog, Microprocessors, Microcontrollers, Hardware Security

Work Experience

Robolabs, LLC (Engineering Division) **May 2025 – Present**
Robotics Systems Engineer *Dublin, CA*

- Developed a real time autonomous mobile robot platform with **EKF** based **multi-sensor fusion (dual GPS, IMU, wheel odometry)** for **3-DOF pose estimation**, **A* grid path planning with collision avoidance**, and a Pure Pursuit controller with adaptive speed ramping, currently investigating **SLAM** based localization approaches and porting the navigation stack to **ROS 2**.
- Engineered a heterogeneous compute pipeline between a **Jetson Nano** (YOLO based object detection at 15 Hz) and an ARM based embedded controller via a custom **CRC32 validated serial protocol**; implemented a high priority receive thread with mutex protected shared state for real time data integrity.
- Built a multi-stage object acquisition pipeline using a **non-blocking FSM** coordinating optical sensors and a 4-stage actuator system; integrated hue based object classification for selective manipulation.

Skills: ROS 2 (actively developing), C++, Sensor Fusion, Autonomous Navigation, Firmware Design/Debug

Biio, Inc. **January 2025 – May 2025**
ASIC & Embedded Design Intern *San Francisco, CA*

- Collaborated on the development of a **SystemVerilog-based feature extraction pipeline** integrating an I2C master-slave communication protocol, sampling interface, filtering logic, and serial output.
- Implemented, a **protocol-accurate finite state machine (FSM)** supporting both read and write operations over the I2C bus. Implemented **edge-detection, clock stretching, and ACK/NACK logic** to ensure timing compliance.
- Designed a synthesizable top-level wrapper integrating streaming data modules (sampfetcher, filteng, featext, and serout) using **clean module interface contracts** and parameterized signal widths to support configurable datapath scaling.
- **Improved RTL verification efficiency by 35%** by developing modular, simulation-ready components with well-defined interfaces and reusable testbenches, laying the foundation for seamless future prototyping and ASIC integration.

Skills: SystemVerilog, Xilinx Vivado, RTL Design, I2C Protocol, FSM Design, Testbench Development, Synthesis & Simulation, ASIC Design

Geopogo **July 2024 – December 2024**
Embedded Software Development Engineer (Contract) *Berkeley, CA*

- Gathered and integrated data for the ZoneQuest platform, streamlining city planning and zoning insights. Leveraged **machine learning algorithms** to refine the user interface, improving overall interface performance by **10%**.
- Worked with hardware-based deep neural network (DNN) algorithms to integrate a Generative AI system into the Magic Leap 2 device, enhancing its real-time processing capabilities and improving user interaction.

Skills: Machine Learning, DNN Optimization, Generative AI, Data Integration

Geopogo **May 2023 – December 2023**
Augmented Reality & Embedded Systems Design Intern *Berkeley, CA*

- Designed and tested interactive 3D models for industrial applications using Unity Engine, focusing on cross-platform deployment and responsive user interfaces.
- Utilized **I2C communication protocols, GPIO (General Purpose Input/Output) handling, and real-time debugging techniques (Serial Debugging and Signal Toggling)** to validate system functionality on embedded hardware.

Skills: Embedded Systems, RTOS Concepts, Multi-threading, Unity Engine, AR/VR Integration, System Debugging, I2C, GPIO

Decibels Lab **May 2021 – July 2021**
RTL Design Intern *Bangalore, India*

- Assembled **RTL** code blocks using **Verilog** on **Vivado**, ensuring high reliability and adherence to technical specifications. Conducted detailed troubleshooting and refinement of firmware for **FPGA-based systems**, enhancing operational efficiency by **20%**; conducted STA on **Vivado**.

Skills: RTL Design, FPGA Development, Xilinx Vivado, Static Timing Analysis, Floorplanning, Synthesis, Testbench Design, Documentation

Computer Systems & Hardware Design Projects

Simulator Enhancements for Krittika - SIMD Units and Network on Chip | *GaTech* **March 2024 - April 2024**

- Extended Krittika to support 11 new **SIMD-based non-linear activation functions** (e.g., GeLU, Swish, Mish, Sigmoid), enabling compatibility with modern DNN models like **AlexNet** and **BERT**.
- Set up a complete **RISC-V simulation environment** using Rocket Core, GNU Toolchain, Spike ISA Simulator, and Proxy Kernel. Developed **C-based** benchmarks, compiled with **RISC-V GCC** using architecture-optimized flags for double-precision workloads.
- Used **cycle-accurate profiling and event-driven simulation** to validate function latency (e.g., Softmax: 1s, Tanh: 94s), improving simulation fidelity for hardware-accelerated deep learning.

Skills: RISC-V, GNU Toolchain, Spike Simulator, C, SIMD Profiling, Deep Learning Accelerators, Linux

16*4 SRAM Array design using 45nm Predictive Technology | *GaTech* **October 2023 - December 2023**

- Designed a **16×4 SRAM array** using **6T SRAM cell** in **Cadence Virtuoso**, integrating row/column decoders, sense amplifiers, wordline drivers, and pre-charge circuits.
- Performed **timing** and **margin analysis** by sweeping cell widths and source voltage (VDD) to evaluate access time, read stability, and write-stability; noticed an increase in read and write margins with an increase in VDD.
- Achieved an array power of 1.2mW at 800mV by applying power optimization and synchronization techniques.

Skills: Cadence Virtuoso, SRAM Design, Analog Circuit Design, Power Optimization Techniques, Timing Analysis

Multi-core Cache Partitioning for Multi-level system with DRAM main memory | *GaTech* **November 2022**

- Built a multi-level cache hierarchy featuring private L1 and shared L2 caches, configurable with varying associativity, line size, cache size, and replacement policies.
- Implemented a DRAM model with **16 banks** and per-bank row buffers, supporting **open-page** and **close-page** policies for realistic memory access timing.
- Enabled detailed performance evaluation of caching strategies, revealing up to **30% reduction** in average memory access time (AMAT) under optimal configurations.

Skills: C++, Cache Architecture, DRAM Modeling, Memory Hierarchy, Replacement Policies

Superscalar Out-of-Order Pipeline with In-Order Commit | *GaTech* **October 2022 - November 2022**

- Implemented a **superscalar processor pipeline** in **C++** with register renaming using a **Register Alias Table (RAT)** and **Re-Order Buffer (ROB)** to maintain precise execution state; supported both **in-order and out-of-order** execution, simulating variable load latencies to model realistic memory behavior.
- Achieved an average CPI of 1.033 across diverse benchmark programs, demonstrating efficient instruction-level parallelism.

Skills: C++, Superscalar Pipeline, Register Renaming (Tomasulo Algorithm), Out-of-Order Execution, Computer Architecture

C++ based driver for ENS160 Sensor using mbed LPC1768 | *GaTech* **October 2022 - November 2022**

- Developed a **C++-based I2C driver** for the mbed LPC1768 board to use the SparkFun Qwiic ENS160 air quality sensor to read live environmental data.
- Interfaced the driver with an LCD display to visualize real-time Air Quality Index (AQI), CO2 levels, and Total Volatile Organic Compounds (TVOC).

Skills: C++, I2C Protocol, Embedded Systems

PACMAN Game using SFML | *GaTech* **September 2022**

- Developed a classic PACMAN game in **C++** using the **SFML graphics library**, featuring dynamic sprite animations, collision detection, and responsive user controls.
- Engineered modular game components such as maze generation, coin/power-up logic, and ghost AI with randomized movement, wall collision handling, screen wrap-around, and timed power-up states for smooth gameplay.

Skills: C++, SFML

Superscalar Pipeline with Data Forwarding and Branch Prediction | *GaTech* **August 2022 - September 2022**

- Engineered a high-performance system with **branch prediction** and **data forwarding** in **C++** by integrating **Always Taken**, **GShare** and **Bimodal** predictors; implemented a **Global History Table (GHT)** for indexing into the **Pattern History Table (PHT)**.
- Achieved a **20% reduction** in branch mispredictions and increased overall CPU performance.

Skills: C++, Branch Prediction, Data Forwarding, Computer Architecture

Technical Skills

Programming Languages: C, C++, Verilog, SystemVerilog, Assembly Language, TCL, SQL, Python
Embedded & Hardware: Jetson Nano, LPC1768, Arduino, Raspberry Pi, ESP8266, ENS160, x86; I2C, SPI, UART
Robotics & Autonomy: ROS 2 (learning), Sensor Fusion, A* Path Planning, PID Control, Object Detection, Autonomous Navigation
Technologies: Git, VS Code, Linux, Cursor AI

Publications

- ***"Ultrasound Image Analysis of Carotid Artery"***
  S S Suryaa, Srinivas Raman S, Ajay Prakash K, Dr. S. Latha, Dr. P. Muthu.
  Accepted at IEEE International Conference of Computing and Communication Technologies (ICCCT), 2025.
  Available: https://ieeexplore.ieee.org/document/11019420.